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Paper Code : OEC-IT701A/OECCS701A/OECDS 701A Operations Research

UPID : 007549

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) In queuing system, _____ refers to those waiting in a queue or receiving service.
a) Service provider b) Client c) Customer d) Patron

(II) When the total demand is equal to the total supply then the transportation problem is said to be:
a) Balanced b) Unbalanced c) Maximization d) Minimization

(III) An activity that does not consume neither any resource nor time is known as:
a) Null activity b) Dummy activity c) Predecessor activity d) Special activity

(IV) A matrix which shows the gains and losses resulting from moves and counter moves is called:
a) Cost matrix b) Pay-off matrix c) Gain matrix d) Loss matrix

(V) A queue is formed when the demand for a service:
a) exceeds the capacity to provide that service.
b) is less than the capacity to provide that service.
c) is equal to the capacity to provide that service.
d) there is no relation between service capacity and queue.

(VI) LPP involving four variables can be solved by:
a) Simplex method b) Graphical method c) Matrix-minima method d) Hungarian method

(VII) An activity is said to be critical activity if:
a) its free float is zero
b) its total float is zero
c) its independent float is zero
d) its time duration is zero

(VIII) The position in the pay-off matrix where the maximin coincides with the minimax is called:
a) Saddle point b) Break even point c) Pivot point d) Key point

(IX) Commonly assumed probability distribution of customer arrival pattern in a queue is:
a) Poisson distribution b) Binomial distribution c) Normal distribution d) Beta distribution

(X) To find initial basic feasible solution of transportation problem the difference of smallest cost and next higher cost of each row and column is related to:
a) N-W corner method
b) Least cost method
c) Matrix minima method
d) Vogel approximation method

(XI) What is expected time value in a PERT problem with to, tm, and tp as 6, 10 and 14 days respectively?

(XII) In a _____ , the amount won by all the winners together is equal to the sum of the amounts lost by all the losers together.
a) Non-zero sum game b) Zero sum game c) Rectangular game d) Equal game

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Construct a network for each of the activities and their precedence relationships are given below: [5]

Activities	A	B	C	D	E	F	G
Predecessor	---	---	A	A	B	C	D, E

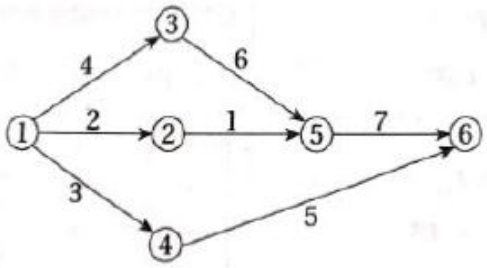
3. Apply maximin and minimax principle to solve the game whose pay-off matrix is given below: [5]

	B1	B2	B3	B4
A1	-5	2	1	6
A2	5	6	4	8
A3	4	0	1	-3

4. In a public telephone booth the arrivals are on average 15 per hour. A call on the average takes 3 minute. If there is just one phone, find the proportion of the time the booth is expected to be idle. [5]

5. Show that the vectors (1,2,3) and (4, -2,7) are linearly independent. [5]

6. For the following network, find the critical path and the expected time to complete the project. [5]



Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. Details of a small project are given below. [15]

i) Draw a network diagram for this project. Find the critical path, the expected project completion time.

ii) What project duration will have 95% confidence of completion?

[Given, $P(Z \leq 1.64) = 0.95$]

Activity	Most Likely	Optimistic	Pessimistic	Immediate Predecessor
A	3	1	7	—
B	6	2	14	*A*
C	3	3	3	*A*
D	10	4	22	*B, C*
E	7	3	15	*B*
F	5	2	14	*D, E*
G	4	4	4	*D*

8. Solve the following pay-off matrix; determine the optimal strategies and the value of the game [15]

$$\begin{array}{c} \\ A \end{array} \begin{array}{c} \begin{array}{cc} B & \end{array} \\ \begin{bmatrix} 5 & 1 \\ 3 & 4 \end{bmatrix} \end{array}$$

9. (a) A two channel waiting line with Poisson arrival has a mean arrival rate of 50 per hour and exponential service with a mean service rate of 75 per hour for each channel. Find the probability that an arrival in the system will have to wait. [7]

(b) An ice-cream company sales one of its types of ice-cream by weight. If the product is not sold on the day it prepared, it can be sold at a loss of 50 paise per pound. But there is an unlimited market for one day ice-cream. On the otherhand, the company makes a profit of Rs. 3.20 on evey pond of ice-cream sold on the day it is prepared. The past daily orders form a distribution $f(x) = 0.2 - 0.0002x$, $0 \leq x \leq 100$ [8]

How many pound of ice-cream should the company prepare every day?

10. Solve the Assignment problem for minimum cost. [15]

Men / Machine	A	B	C	D
M_1	18	26	17	11
M_2	13	28	14	26
M_3	38	19	18	15
M_4	19	26	24	10

11. Solve the following rectangular game by graphical method. [15]

$$\begin{array}{c} \\ \text{Player A} \end{array} \begin{array}{c} \begin{array}{cccc} \text{Player B} & & & \end{array} \\ \begin{bmatrix} 1 & 0 & 4 & 1 \\ -1 & 1 & -2 & 5 \end{bmatrix} \end{array}$$